

for the current state of Selling engine :
COMMON – NUMERICAL – SELLING – KERNEL = PROVEN – EXACT
and for Selling – Noether bridge

$$\Phi_{N \rightarrow \text{Sell}} : (f, \rho) \mapsto Q, \quad Q_{ij} = \sum_{k,l} f_{ijkl} \rho^{kl}.$$

$$\text{here} : f \in \text{Sym}^4(V), \quad \rho \in \text{Sym}^2(V), \quad Q \in \text{Sym}^2(V).$$

$$\text{for } g \in GL(3) \quad g \cdot f = f \circ g^{-1}, \quad g \cdot \rho = g \rho g^T,$$

$$\text{the contraction gives exactly} : \quad \Phi(g \cdot f, g \cdot \rho) = g^{-T} \Phi(f, \rho) g^{-1}.$$

posing $B = g^{-1}$ *that become precisely the Selling base change law :*

$$Q \mapsto B^T Q B.$$